

Physics 5C Discussion (Worksheet 7)

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In this worksheet, we will analyze circuits using Kirchhoff's circuit laws. The junction rule says that the total current entering a junction equals the total current leaving the junction,

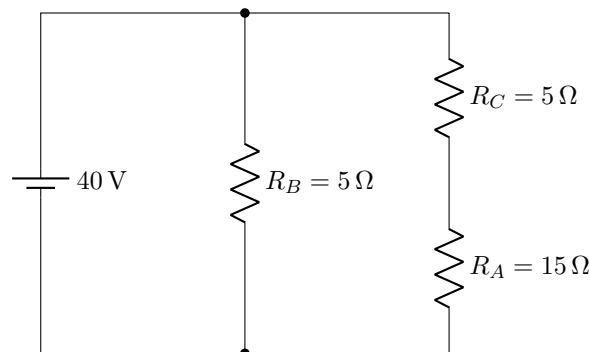
$$\sum I_{\text{in}} = \sum I_{\text{out}}.$$

The loop rule says that the sum of all potential changes around any closed loop is zero,

$$\sum \Delta V = 0.$$

For all problems, assume that the batteries and wires have negligible resistance.

Problem 1 Jinyan has built the same circuit that appeared in Problem 2 of Worksheet 6, redrawn below as a standard circuit diagram. This time, do not use equivalent resistance. Analyze the circuit using the junction rule and loop rule.

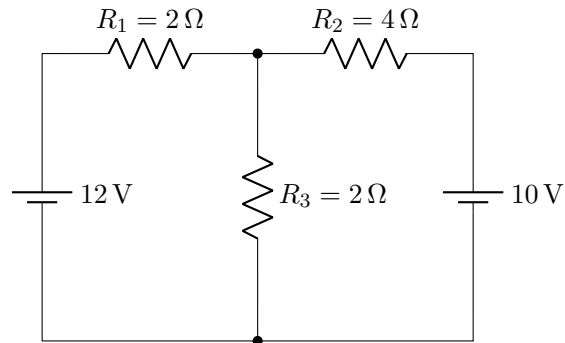


Part (a) Choose a direction for the current through the battery and for the current in each resistor branch. Write a junction-rule equation relating these currents.

Part (b) Apply the loop rule to two suitable loops and use the resulting equations, together with the junction rule, to find the current through the battery and the current through each resistor.

Part (c) Find the voltage drop across resistor A .

Problem 2 Now consider the multi-loop circuit shown below. There are two batteries in different branches of the circuit, and the upper terminal of each battery is positive. Find the current through each resistor using Kirchhoff's circuit laws.



Part (a) Choose a direction for each of the three unknown currents I_1 , I_2 , and I_3 , where I_1 , I_2 , and I_3 are the currents through R_1 , R_2 , and R_3 , respectively. Apply the junction rule at the top central junction.

Part (b) Determine how many equations are needed to solve for the three unknown currents. Choose appropriate loops, apply the loop rule, and write a system of equations that contains enough different information to solve the problem.

Part (c) Solve the system of equations and find I_1 , I_2 , and I_3 . If any current is negative, explain what the negative sign means.

Part (d) Find the voltage drops across the resistors and verify the loop rule for the loops used in part (b).

Problem 1

Let us first choose current directions. I will call the current traveling upward through the battery I_0 . At the top junction, I assume that this current splits into a downward current I_B through resistor B and a downward current I_{CA} through the branch containing resistors C and A . These directions are guesses that allow us to write equations consistently. If a current turns out to be negative, that only means the real current points opposite to the direction that we guessed.

Part (a) At the top junction, I_0 enters, while I_B and I_{CA} leave. The junction rule therefore gives

$$I_0 = I_B + I_{CA} . \quad (1)$$

There are three unknown currents, so we need three equations that provide three different pieces of information. Equation (1) is one of them. Applying the junction rule at the bottom junction would give the same information written in reverse, so that would not help us determine the three unknowns. We need two more useful equations, which we obtain from the loop rule.

Part (b) Before writing the loop equations, let us establish the signs. When we cross a battery from its negative terminal to its positive terminal, the potential change is positive. That is, the battery is supplying some voltage. When we move through a resistor in the same direction as the current, the potential change is negative because there is a voltage drop $\Delta V = IR$. That is, the resistor is consuming some voltage supplied by the battery. Moving through a resistor opposite to the current instead gives a positive potential change.

For the left loop, start at the negative terminal of the battery, travel upward through the battery, then travel downward through resistor B . We cross the battery from negative to positive, giving $+40\text{ V}$, and we move through B in the direction of I_B , giving a drop $-I_B R_B$. The loop rule gives

$$0 = V_{\text{battery}} - V_B = V_{\text{battery}} - I_B R_B = (40\text{ V}) - I_B(5\ \Omega) . \quad (2)$$

Therefore,

$$I_B = \frac{40\text{ V}}{5\ \Omega} = 8\text{ A} .$$

For the outer loop, start at the negative terminal of the battery, travel upward through the battery, then travel down the branch containing C and A . The same current I_{CA} passes through both C and A because there is no junction between them. The loop rule gives

$$0 = V_{\text{battery}} - V_C - V_A = V_{\text{battery}} - I_{CA} R_C - I_{CA} R_A = (40\text{ V}) - I_{CA}(5\ \Omega) - I_{CA}(15\ \Omega) . \quad (3)$$

Thus, rearranging, we find

$$I_{CA} = \frac{40\text{ V}}{20\ \Omega} = 2\text{ A} .$$

You might have noticed that there is also a loop on the right in the circuit diagram: If we start from the bottom junction and go clockwise, we first go through resistor B in the direction opposite to our guessed current direction, then we pass the top junction, make a right turn, and go through C and A aligned with our guessed direction of current. So we also have a loop rule here

$$0 = V_B - V_C - V_A = I_B R_B - I_{CA} R_C - I_{CA} R_A = I_B(5\Omega) - I_{CA}(5\Omega) - I_{CA}(15\Omega). \quad (4)$$

In fact, we only need to use two of the three equations (2), (3) and (4) obtained from the loop rule to solve for the two unknowns I_B and I_{CA} . You can verify that, indeed, a combination of any two of them can do the job. Some combinations are easier (such as the one we did with equations (2) and (3)), but some others are a little more challenging.

Now use the junction equation (1),

$$I_0 = I_B + I_{CA} = 8\text{ A} + 2\text{ A} = 10\text{ A}.$$

Therefore, the current through resistor B is 8 A, the current through both C and A is 2 A, and the current through the battery is 10 A. All three results are positive, so the actual currents point in the directions that we assumed.

Notice that we did not need to calculate an equivalent resistance. The junction rule told us how the current splits, while the loop rule required the potential gained from the battery to equal the potential dropped across the resistors in each closed path.

Part (c) Since the current through resistor A is $I_A = I_{CA} = 2\text{ A}$, Ohm's law gives

$$V_A = I_A R_A = (2\text{ A})(15\Omega) = 30\text{ V}.$$

As a quick check, the other 10 V is dropped across resistor C , because

$$I_C R_C = (2\text{ A})(5\Omega) = 10\text{ V}.$$

The two voltage drops add to the battery voltage, $30\text{ V} + 10\text{ V} = 40\text{ V}$, as required by the loop rule.

Problem 2

The equivalent-resistance approach is not useful for the entire circuit in this problem. Resistors R_1 and R_2 are not in series because their common connection is also attached to R_3 , creating a junction. No two resistors are in parallel because no pair is connected to the same two junctions. More importantly, the two batteries are located in different branches, so there is no single battery voltage that we can simply divide by a total equivalent resistance to obtain all the branch currents. We need the junction rule and loop rule to keep track of both power sources and all three currents.

Part (a) Focusing on the top central junction: I will assume that I_1 travels from the left battery through R_1 toward the top central junction. I will assume that I_2 travels

from the right battery through R_2 toward the same junction. Finally, I will assume that I_3 travels downward through R_3 . At the top central junction, I_1 and I_2 enter while I_3 leaves, so the junction rule gives

$$I_1 + I_2 = I_3. \quad (1)$$

These directions are only our starting guesses. We do not need to know the actual directions before solving the circuit. The signs of the final answers will tell us whether the guesses were correct.

Part (b) We have three unknowns, I_1 , I_2 , and I_3 , so we need three equations containing three different pieces of information. The junction equation (1) gives one. For the other two, choose the small loop on the left and the small loop on the right. Of course, we could have also used the big outer loop containing both power sources and R_1 and R_2 . These three loops give three equations, but we only need two of them, and you can check that any two of them suffice.

For the left loop, begin at the negative terminal of the 12 V battery and travel clockwise. We cross the battery from negative to positive, travel through R_1 in the direction of I_1 , and then travel downward through R_3 in the direction of I_3 . Thus,

$$0 = V_{\text{left batt.}} - I_1 R_1 - I_3 R_3 = 12 - 2I_1 - 2I_3. \quad (2)$$

Here the numerical values are understood to carry the appropriate units.

For the right loop, begin at the negative terminal of the 10 V battery and travel counterclockwise. We cross that battery from negative to positive, travel through R_2 from right to left in the direction of I_2 , and then travel downward through R_3 in the direction of I_3 . Therefore,

$$0 = V_{\text{right batt.}} - I_2 R_2 - I_3 R_3 = 10 - 4I_2 - 2I_3. \quad (3)$$

Equations (1), (2), and (3) are enough to solve the three unknown currents. Again, we could also write a loop equation around the large outside loop, but it would only repeat information already contained in equations (2) and (3). Similarly, applying the junction rule at the bottom central junction would reproduce equation (1). When counting equations, we should count only equations that tell us something new. A reliable strategy is to use all but one of the junction equations and then add loop equations until the number of useful equations matches the number of unknown currents.

Part (c) Now it is just a matter of solving a system of equations,

$$\begin{cases} I_1 + I_2 = I_3 \\ 12 - 2I_1 - 2I_3 = 0 \\ 10 - 4I_2 - 2I_3 = 0 \end{cases}.$$

There are many ways to solve this system of equations. My attempt starts by summing the first two equations to obtain

$$2I_1 + 2I_2 + 12 - 2I_1 - 2I_3 = 2I_3 + 0$$

$$2I_2 + 12 - 2I_3 = 2I_3$$

$$I_2 + 6 = 2I_3$$

Then we can combine this with the third equation to isolate I_2 ,

$$10 - 4I_2 - I_2 - 6 = 0 ,$$

from which we find $I_2 = 0.8 \text{ A}$. Then using the third equation with $I_2 = 0.8 \text{ A}$ we find $I_3 = 3.4 \text{ A}$, and lastly the first equation with the found values of I_2 and I_3 gives $I_1 = 2.6 \text{ A}$. We thus have found

$$I_1 = 2.6 \text{ A} , \quad I_2 = 0.8 \text{ A} , \quad I_3 = 3.4 \text{ A} .$$

All three currents are positive. Therefore, our assumed directions were correct: I_1 travels from the left toward the central junction, I_2 travels from the right toward the central junction, and I_3 travels downward through R_3 . If one of the answers had been negative, we would keep its magnitude but reverse its direction relative to our original arrow.

Part (d) The current through each resistor has been found in part (c), so we can use Ohm's law to find the corresponding voltage drops,

$$\begin{aligned} V_1 &= I_1 R_1 = (2.6 \text{ A})(2 \Omega) = 5.2 \text{ V} , \\ V_2 &= I_2 R_2 = (0.8 \text{ A})(4 \Omega) = 3.2 \text{ V} , \\ V_3 &= I_3 R_3 = (3.4 \text{ A})(2 \Omega) = 6.8 \text{ V} . \end{aligned}$$

These are voltage drops when we travel through the resistors in the directions of the currents that we found in part (c). If we travel through a resistor opposite to the direction of its current, the corresponding potential change has the opposite sign.

Now we verify the loop rule for the two loops used in part (b). For the left loop, begin at the negative terminal of the 12 V battery and travel clockwise. The battery gives a potential increase of 12 V , while resistors R_1 and R_3 give voltage drops of 5.2 V and 6.8 V . For the right loop, begin at the negative terminal of the 10 V battery and travel counterclockwise. That battery gives a potential increase of 10 V , while resistors R_2 and R_3 give voltage drops of 3.2 V and 6.8 V . Therefore,

$$\begin{aligned} \sum \Delta V_{\text{left loop}} &= V_{\text{left batt.}} - V_1 - V_3 = 12 \text{ V} - 5.2 \text{ V} - 6.8 \text{ V} = 0 \text{ V} , \\ \sum \Delta V_{\text{right loop}} &= V_{\text{right batt.}} - V_2 - V_3 = 10 \text{ V} - 3.2 \text{ V} - 6.8 \text{ V} = 0 \text{ V} . \end{aligned}$$

The sum of the potential changes is zero for both loops, as required by Kirchhoff's loop rule.



Visited University of Virginia in August, 2026.